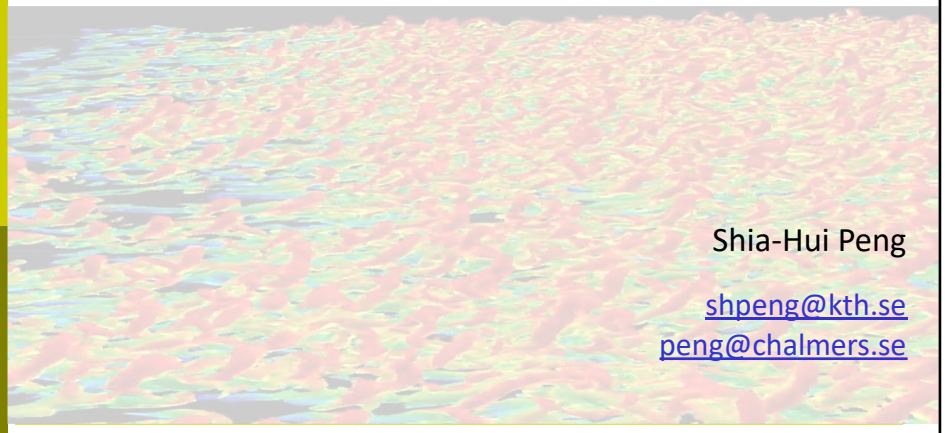
Introduction about

Computational Fluid Dynamics for Built Environment

(*Simulations for Fluid and Climate Technology*)



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Outlines

- Conceptual introduction about CFD
  - What is CFD?
  - Where is CFD used (some examples)?
- Some important aspects in CFD
  - Numerical methods
  - Boundary conditions
  - **Turbulence (flow physics) modelling**
- CFD for indoor environment (cf. reference document)

What is CFD (Computational Fluid Dynamics)?

- Computer-based modelling & simulation for analyzing fluid flow and related transport phenomena (e.g., heat and mass transfer).
- Solving *numerically* for a system of Partial Differential Equations (PDEs) that govern fluid motions, and related heat / mass transfer, discretized in algebra forms on a grid and programmed in a computer software.
- CFD is commonly applied in engineering for numerical analysis of, typically,
  - Flow-system performance, problem diagnosis, design optimization …
  - Heat and mass transfer
  - Fluid-structure interaction
  - Acoustic noise analysis …. and many more ….
- CFD applications to built environment started in 1970's, and having been facilitated greatly by the development of computing facilities
- **CFD simulations are usually verified by**
  - Analytical solutions (of simple flow problems)
  - Experimental measurements
  - Cross comparisons of results computed with different CFD codes

Where is CFD Used?

- Theoretically, for any transport phenomena present in fluid motions and in flow systems - in nature or created for engineering use
- Typical applications include, among others,
  - Oceanic circulation, metrological and climatic predictions
  - Power generation (hydro- and wind- power plant …)
  - Automotive industry and appliances manufacturing (cooling, dehumidifying …)
  - Chemical processing
  - Oil and gas industry
  - Hydraulics, maritime & underwater applications
  - Sport (e.g., drag reduction of swimming, running, canoeing, …)
  - Aerospace engineering
  - **Built environment and indoor climate**
  - ***Medical engineering***
  - ……

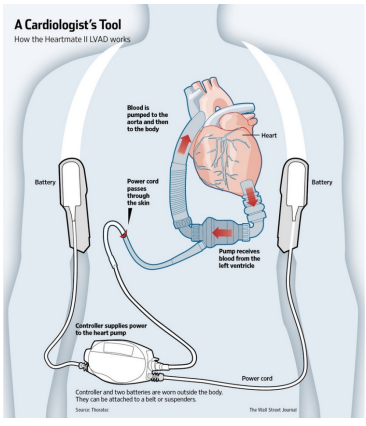
  Serving optimized designs, problem diagnosis, and analysis. Usually is more efficient than experimental measurements.
- Practically, use of CFD may be limited by computing capabilities
- Accuracy of CFD affected by physical modelling and many other factors

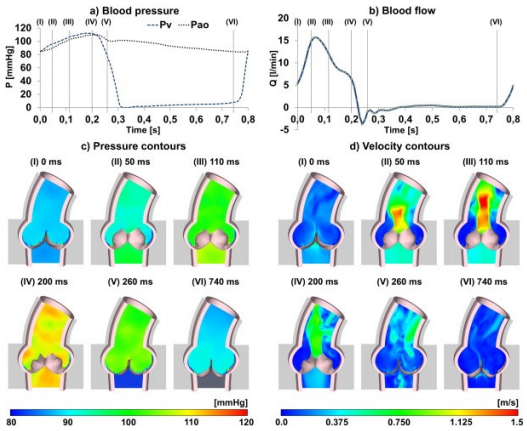
Example – CFD simulation for indoor environment

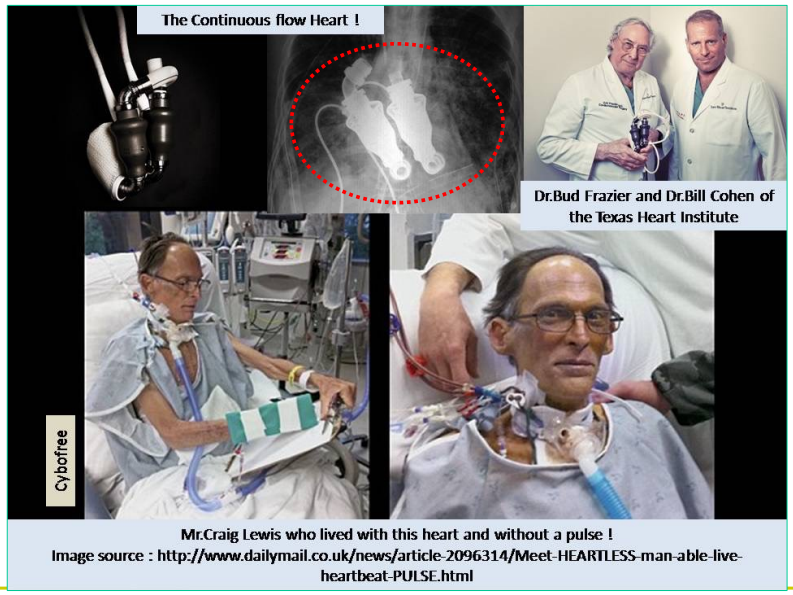


Animations taken from YouTube only for demo.

Example – Simulation of blood flow (heart pump)

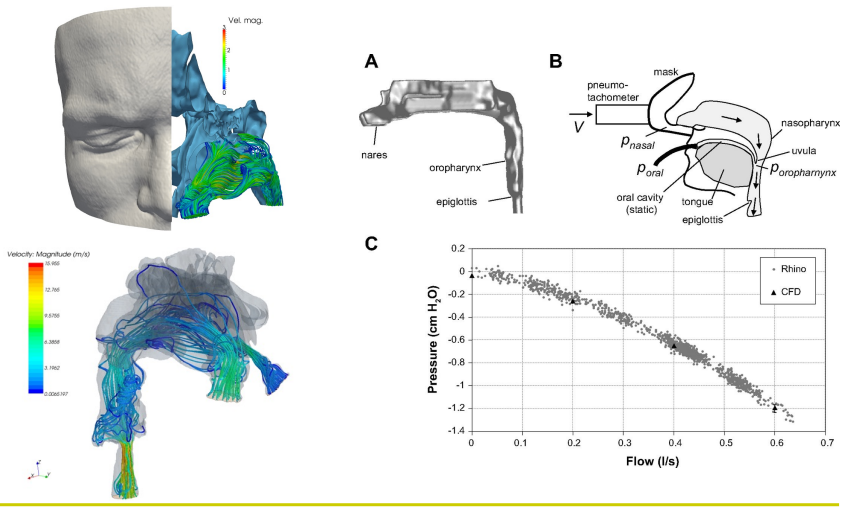






Mr.Craig Lewis who lived with this heart and without a pulse !
Image source : http://www.dailymail.co.uk/news/article-2096314/Meet-HEARTLESS-man-able-live-heartbeat-PULSE.html

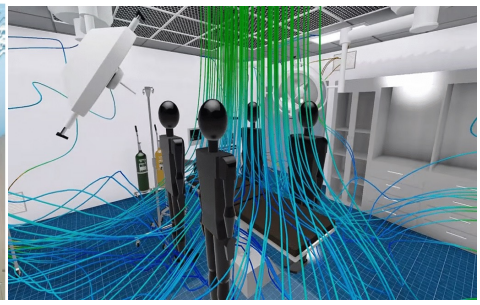
Example – Analysis of nasal airflow



Example – Simulations of clean airflow in surgery theater

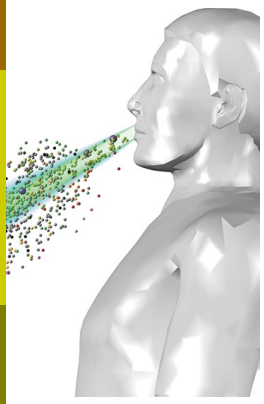


Ventilation in operating rooms

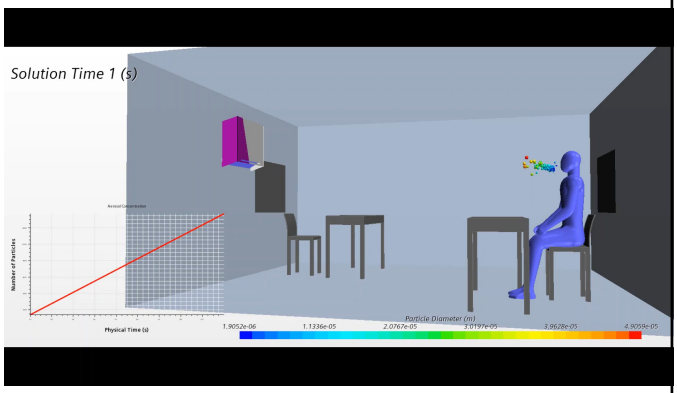


Airflow distribution designed to prevent transmissions of surgical site infections

Example - Modelling of ventilating sneeze-generated droplets

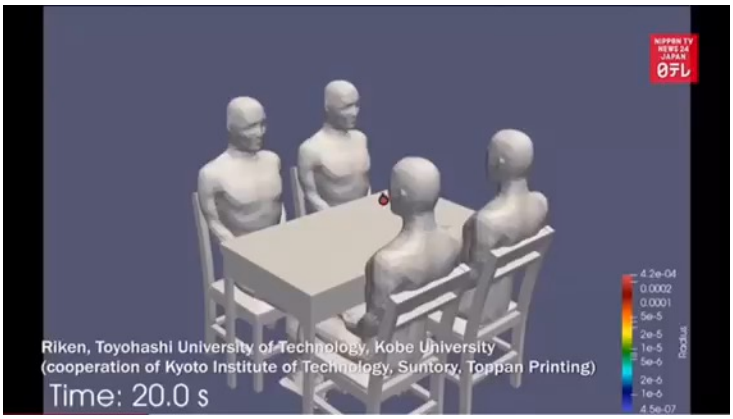


Droplets from sneeze



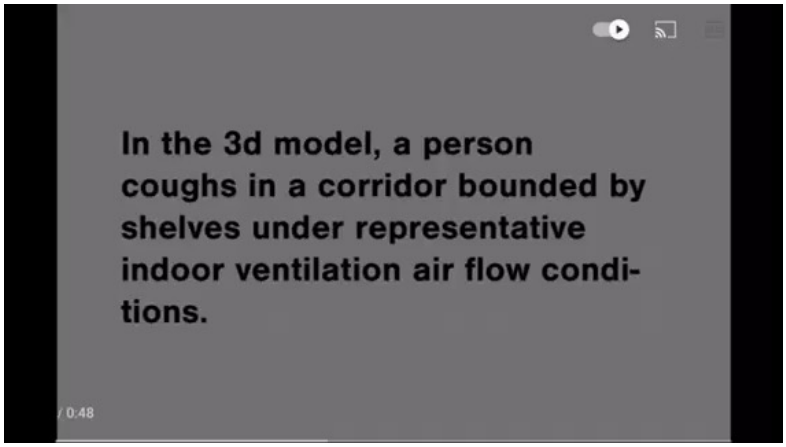
Number and distribution of sneeze-generated droplets changing with time in a ventilated room

Example – Modelling of Indoor Air Quality (IAQ)



CFD analysis of droplets transport in a meeting room

Example – Modelling of dispersion of aerosol cloud in a ventilated workplace



Transmission of cough-induced droplets in a landscape-type working place

CFD is used in engineering for ……

- Conceptual studies of, e.g., fluid-flow systems
- Design & design optimization
- Off-design analysis
- Problem diagnosis & analysis
- Provision with much detailed flow information
- Supplement to experimental measurements

BUT

- Limited by computing resources, particularly, for large-scale and/or high-Re flows
- Accuracy of CFD affected by, typically, ***physical modelling***, ***numerical settings & algorithms*** *incorporated in CFD solver*, ***boundary conditions***, *skill and experience of users …*

Key issues in CFD simulations (1)

- **Modelling of flow physics**: Equation systems (mathematical modelling) governing fluid motions & inherent transport phenomena
  - **Incompressible** **or** Compressible
  - **Newtonian** or Non-Newtonian fluid
  - **Steady** or **Unsteady**
  - **Viscous** or Invicid
  - Laminar or **Turbulent**
  - **Heat & mass transfer**
  - Multiphase, chemical reaction

Indoor airflow is usually incompressible turbulent flows

Furthermore, applications coupled with modelling of, e.g.,
- Free surface (e.g., flow around ship, offshore wind farm)
- Multi-phase flows with or without phase change (e.g., nucleate boiling)
- Fluid-structure interaction (e.g., flow-induced oscillations)
- Acoustic noise (e.g., aircraft noise, fan noise …)

Key issues in CFD simulations (2)

- **Numerical schemes & algorithms:** Closely associated to computational accuracy and efficiency
  - Grid generation & resolution
    (Structured, unstructured grid …)
  - Spatial and temporal discretization (numerical schemes)
    Finite element, finite difference, or finite volume method …
  - Solution algorithms
  - … …
- **Boundary conditions:** Problem-dependent, greatly impacting on computational accuracy
  - Solid wall boundaries
  - Oscillating/moving walls
  - Walls with heat transfer
  - In/out-flow boundaries
  - Symmetric, periodic boundaries
  - Near-wall treatment (wall functions …)
  - …

Typical boundary conditions in CFD analysis of built-environment

- **Solid wall boundaries**
  - Velocity components: u = v = w = 0 at the wall (no-slip)
  - *Temperature for walls with heat transfer*: T = *constant* (Dirichlet or 1st-type); $(\partial T/\partial \boldsymbol{n})_{wall}$ = constant (Neumann or 2nd-type)
  - Temperature for adiabatic wall: $(\partial T/\partial n)_{wall} = 0$ (no heat transfer)
  - Turbulent quantities (k = 0 at wall ….)
- **Inlet boundaries**
  - Velocity components & temperature: known, or given usually from measurements
  - Turbulent quantities, usually estimated or measured
- **Outlet boundaries**
  - Usually Neumann condition $\partial f/\partial \boldsymbol{n} = 0$ (BUT Note: mass and energy conservation must be ensured for the flow system)
- **Symmetric boundary**
  - Usually Neumann condition $\partial f/\partial \boldsymbol{n} = 0$ (no flux passing through the boundary)

NOTE: **n** is the direction normal to the boundary.

Questions & Discussion:

1. What is CFD in brief? How is usually a CFD simulation verified?

   Ref. Slide 3

2. What are the three key "elements" in CFD that may affect significantly simulation accuracy and efficiency?

   Ref. Slides 14-15

3. Why are boundary conditions needed in CFD simulations?

   To ensure unique solution of a specific flow problem that is well posed

Introduction about Grid Generation in CFD

(*Given in other lectures ……?*)

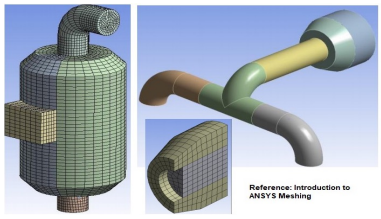
- Grid design and grid quality
- Grid adaptation and refinement

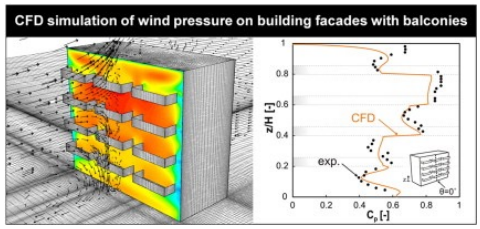
Doc. Notes Grid

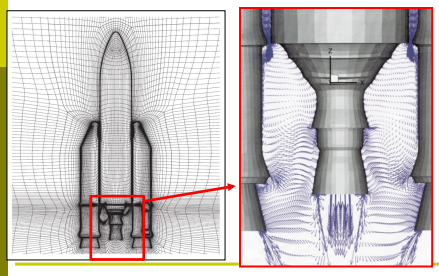
Discretized equations are solved at each node of a grid, which
- *covers the whole computational domain*
- *represents appropriately the geometries (boundaries)*
- *Many software for grid generation (ICEM-CFD, Pointwise ….)*

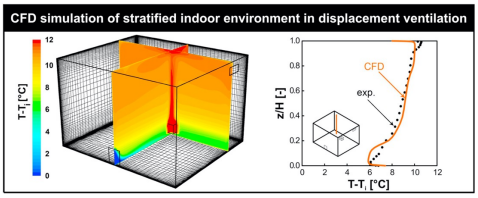
Usually, a refined grid resolution is required in regions with large gradients of flow variables → "**grid sensitivity**" study
NOTE: velocity gradients → shear (e.g., near the wall)

- CFD simulation is conducted with a grid that maps appropriately the geometric configuration and is adapted to local flow features of the flow system.









Introduction about Turbulence Modelling

- **RANS** (**R**eynolds **A**veraged **N**avier-**S**tokes Equations
- **URANS** (**U**nsteady RANS)
- **LES** (**L**arge **E**ddy **S**imulation)
- **SGS** (**S**ub-**G**rid **S**cale)
- **DNS** (**D**irect **N**umerical **S**imulation)

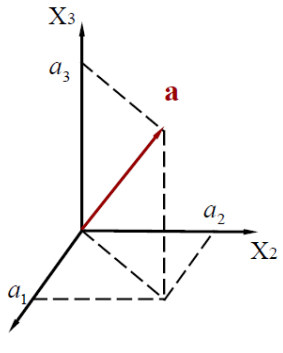
Basics of Cartesian tensors (1)

In a Cartesian coordinate, a vector can be expressed by three ordered scalars

$$\mathbf{a} = a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_3 = \sum_{i=1}^{3} a_i\mathbf{e}_i$$

Summation convention $\mathbf{a} = a_i\mathbf{e}_i$

$\mathbf{e_1}$, $\mathbf{e_2}$ and $\mathbf{e_3}$ are the basis vectors of the Cartesian coordinate system (X_1, X_2, X_3), respectively.



The vector, **a**, can be equivalently expressed with a **first-rank tensor**, a_i, viz.

$$a_i = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}.$$

NOTE: a scalar quantity can be viewed as a **zeroth-rank tensor**.

Basics of Cartesian tensors (2)

Second-rank tensor, a_{ij}.

For example, the **Kronecker delta**, δ_{ij}:

$$\mathbf{e}_i \bullet \mathbf{e}_j = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} = \delta_{ij} \quad \text{or} \quad \delta_{ij} = \begin{cases} 0, & i \neq j, \\ 1, & i = j. \end{cases}$$

A tensor of 2nd-rank can also be expressed with its components in a matrix, e.g., $\delta_{ij} = \mathrm{I} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

For any 2nd-rank tensor, T_{ij}, $T_{ij} = \begin{pmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{pmatrix}$

Note: Levi-Civita's tensor, ε_{ijk} (permutation tensor), is another special tensor.

Basics of Cartesian tensors (3)

Transpose: $(T_{ij})^T = T_{ji}.$

Symmetric: $D_{ij} = D_{ji}$

Anti-symmetric: $R_{ij} = -R_{ji}$

Contraction: $T_{ii} = T_{11} + T_{22} + T_{33}$

Example:

NOTE: i = 1, 2, 3 and j = 1, 2 , 3

$$t_j = n_i T_{ij}$$

Summation about i (= 1, 2 and 3):

$$t_j = n_i T_{ij} = n_1 T_{1j} + n_2 T_{2j} + n_3 T_{3j}.$$

Basics of Cartesian tensors (4)

Example (cont.):

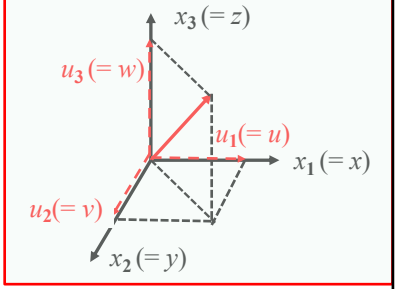
Write down t_1, t_2 and t_3 …. By setting j = 1, 2 and 3, respectively:

$$t_j = n_i T_{ij} = n_1 T_{1j} + n_2 T_{2j} + n_3 T_{3j}.$$

$$t_1 = n_1 T_{11} + n_2 T_{21} + n_3 T_{31}$$

$$t_2 = n_1 T_{12} + n_2 T_{22} + n_3 T_{32}$$

$$t_3 = n_1 T_{13} + n_2 T_{23} + n_3 T_{33}$$



Similarly for the continuity equation:

$$\frac{\partial u_i}{\partial x_i} = 0 \Longrightarrow \frac{\partial u_1}{\partial x_1} + \frac{\partial u_2}{\partial x_2} + \frac{\partial u_3}{\partial x_3} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

From laminar flow to turbulent flow

Reynold Number, Re, as indicator of turbulent flow

After Osborne Reynolds (1842-1912)

$$\mathrm{Re} = \frac{UL}{\nu} = \frac{\rho UL}{\mu} \sim \frac{\textbf{Inertial force}}{\textbf{Viscous force}}$$

U – Flow velocity, $\left[\frac{\mathrm{m}}{\mathrm{s}}\right]$

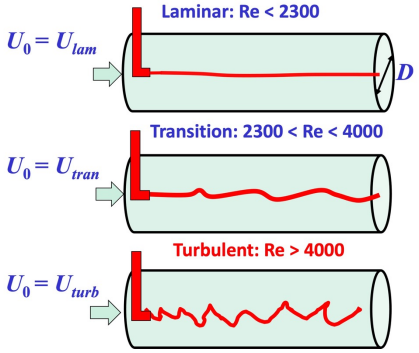
L – Characteristic length scale of flow configuration, [m]

ν – Kinematic viscosity of fluid, $\nu = \frac{\mu}{\rho}$, $\left[\frac{\mathrm{m}^2}{\mathrm{s}}\right]$

μ – Dynamic viscosity of fluid, $\left[\frac{\mathrm{kg}}{\mathrm{m{\cdot}s}} = \frac{\mathrm{N{\cdot}s}}{\mathrm{m}^2}\right]$

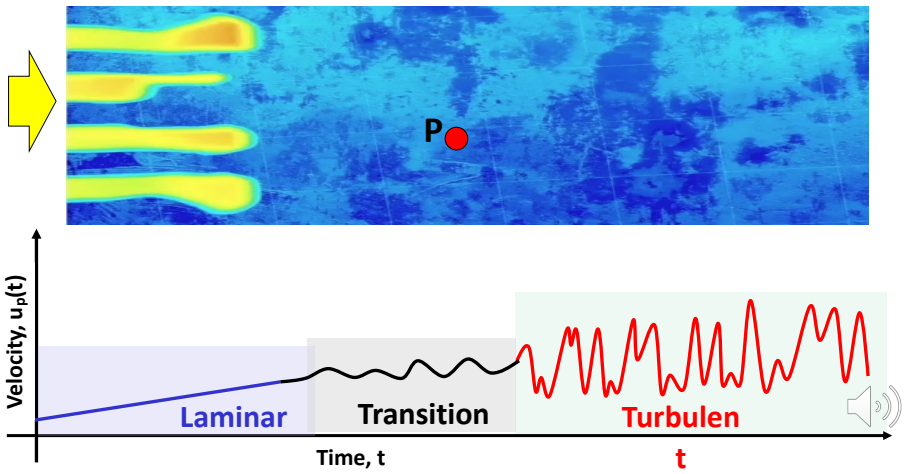
ρ – Density of fluid, $\left[\frac{\mathrm{kg}}{\mathrm{m}^3}\right]$

Pipe flow: $\mathrm{Re} = \frac{U_0 D}{\nu}$. $U_{lam} < U_{tran} < U_{turb}$



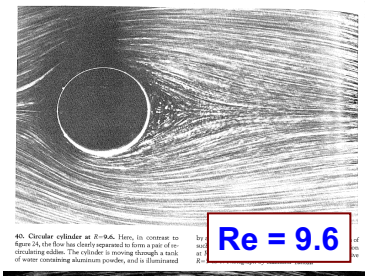
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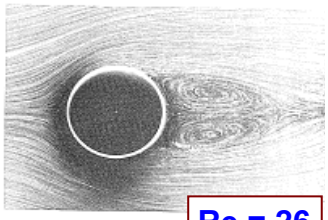
Evolution from laminar to turbulent flow with increasing Re number

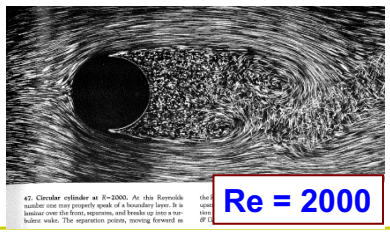


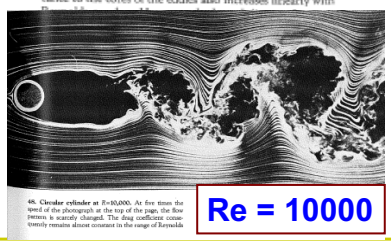
26

From laminar to turbulence: Flow around a cylinder









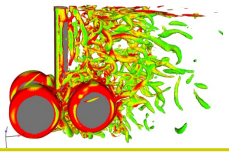
Nature of Turbulence

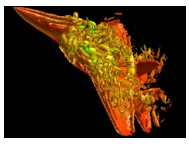
- Chaotic, irregular and unsteady
- Large Reynolds number (large ratio of time scales)
- Swirling flow structures (eddies, vortices…..)
- Three-dimensional fluctuations
- Diffusive
- Dissipative

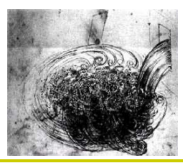
Turbulent fluid motion is inherently rotational

Not a property of fluid, but of continuum flows!









Strain Rate and Vorticity

(Fundamental concepts in defining flow properties and physics)

Symmetric derivative (Strain rate tensor)

$$S_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)$$

Antisymmetric derivative (vorticity tensor)

$$\Omega_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}\right)$$

Stemmed from velocity gradients

Rate of strain (magnitude)

$$|S| = \sqrt{2S_{ij}S_{ij}}$$

Vorticity (magnitude)

$$\Omega = \sqrt{2\Omega_{ij}\Omega_{ij}} = |\nabla \times \mathbf{v}|$$

The strain rate tensor, S_{ij}

$$S_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)$$

$i, j = 1, 2, 3$ (in the x-, y-, z - direction, with velocity components, u_1, u_2 and u_3, or u, v and w)

$$S_{11} = \frac{\partial u_1}{\partial x_1}, S_{22} = \frac{\partial u_2}{\partial x_2}, S_{33} = \frac{\partial u_3}{\partial x_3}$$

$$S_{12} = \frac{1}{2}\left(\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1}\right) = S_{21}, S_{13} = \frac{1}{2}\left(\frac{\partial u_1}{\partial x_3} + \frac{\partial u_3}{\partial x_1}\right) = S_{31}, S_{23} = \frac{1}{2}\left(\frac{\partial u_2}{\partial x_3} + \frac{\partial u_3}{\partial x_2}\right) = S_{32}$$

$$\frac{\partial u_i}{\partial x_i} = 0 \Rightarrow \frac{\partial u_1}{\partial x_1} + \frac{\partial u_2}{\partial x_2} + \frac{\partial u_3}{\partial x_3} = 0$$

Exercise:

Do the same for the **vorticity tensor**, Ω_{ij} given on slide 29, as for S_{ij} above?

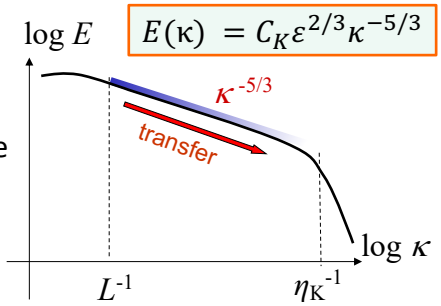
Turbulence Energy Cascade

- Fluid motions are driven by mechanical forces. With increasing Reynolds number and being subjected to large strain rate $|S|$, laminar flow breaks down into large-scale structures (eddies) of high vorticity Ω and exhibiting quasi-random (turbulent) characteristics.
- Large and energy-bearing eddies (of characteristic size, L) progressively break down into even smaller vortices due to nonlinear interactions.
- Turbulence energy is transferred from larger to smaller eddies through a cascading process until it is dissipated into heat due to viscosity at the Kolmogorov scale (smallest eddies).
- The cascade follows Kolmogorov's power law (inertial subrange).

→ Turbulence is non-linear multi-scale phenomena of flow, not of the fluid!

The (-5/3) Power Law for the **inertial subrange**

$E(\kappa) = C_K \varepsilon^{2/3} \kappa^{-5/3}$



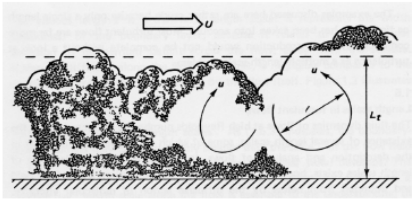
E: turbulence energy in wave-space
ε: Rate of energy dissipation
κ: Wave number, and $\kappa = 2\pi/l$
(l: length scales)

*Energy-carrying structures have larger length scales (or smaller wave numbers); Turbulence energy is transferred from large to small scales and dissipated into heat by the **smallest** structures*

Length scale of viscous dissipation, η_K (the **smallest** Kolmogorov length scale):

$$\eta_K = \nu^{3/4} \varepsilon^{-1/4} \quad \Rightarrow \quad \frac{L}{\eta_K} \approx \mathrm{Re}^{3/4}$$

Turbulence Scales



Large eddies in a turbulent boundary layer (Tennekes and Lumley, 1972)

$\mathcal{L} \sim L_t$ = boundary layer thickness

- Velocity fluctuation, u
- Length (eddy size) scale, $\mathcal{L}$
- Time scale, $t \sim \mathcal{L}/u$
- Turbulent Reynolds number, $Re = u\,\mathcal{L}/\nu$
- Turbulent kinetic energy, $k \sim 3u^2/2$
- Dissipation rate: $\varepsilon \sim u^3/\mathcal{L}$
- Kolmogorov scales:
  - ✓ Time: $\tau_K = (\nu/\varepsilon)^{1/2}$
  - ✓ Length: $\eta_K = (\nu^3/\varepsilon)^{1/4}$
  - ✓ Velocity: $\upsilon_K = (\nu\,\varepsilon)^{1/4}$

Dimensional analysis

$u \sim \left[\frac{m}{s}\right]$; $\mathcal{L} \sim$ [m]; $\varepsilon \sim u^3/\mathcal{L} \sim [m^2/s^3]$, $\nu \sim [m^2/s]$

For the Kolmogorov length scale, which has unit [m], let

$\eta_K = \varepsilon^a \nu^b \sim [m] \Rightarrow [m^2/s^3]^a\,[m^2/s]^b \sim [m]$

$\Rightarrow$ $2a+2b=1$, $3a+b=0$ $\Rightarrow$ $a=-1/4$, $b=3/4$

$\Rightarrow \eta_K = \varepsilon^{-1/4}\nu^{3/4} = \left(\nu^3/\varepsilon\right)^{1/4}$

Exercise:

Try to do the same for the Kolmogorov time scale, τ_K (~ [s]), and velocity scale, υ_K (~ [m/s]).

- Velocity fluctuation, u
- Length (eddy size) scale, $\mathcal{L}$
- Time scale, $t \sim \mathcal{L}/u$
- Turbulent Reynolds number, $Re = u\,\mathcal{L}/\nu$
- Turbulent kinetic energy, $k \sim 3u^2/2$
- Dissipation rate: $\varepsilon \sim u^3/\mathcal{L}$
- Kolmogorov scales:
  - ✓ Time: $\tau_K = (\nu/\varepsilon)^{1/2}$
  - ✓ Length: $\eta_K = (\nu^3/\varepsilon)^{1/4}$
  - ✓ Velocity: $\upsilon_K = (\nu\,\varepsilon)^{1/4}$

Exercise:

(1) With $u \sim \left[\frac{m}{s}\right]$; $\mathcal{L} \sim$ **[m]**, and $\nu \sim$ **[m²/s]**,

using dimensional analysis, show that the Kolmogorov velocity scale can be formulated with $\upsilon_K = (\nu\ \varepsilon)^{1/4}$. (Note $\varepsilon \sim u^3/\mathcal{L}$)

(2) Turbulent Reynolds number is defined by $Re = Re_T = u\,\mathcal{L}/\nu$, referring to the relations on the right-hand-side panel, show that a relation between the length scale of large turbulent structure, $\mathcal{L}$, and the smallest Kolmogorov length scale, η_K, holds as $\mathcal{L}/\eta_K = Re^{3/4}$.

Note: ε is defined by $\varepsilon = u^3/\mathcal{L}$, u and $\mathcal{L}$ are respectively, the velocity and length scale of large turbulent structure.

- Velocity fluctuation, u
- Length (eddy size) scale, $\mathcal{L}$
- Time scale, $t \sim \mathcal{L}/u$
- Turbulent Reynolds number, $Re = u\,\mathcal{L}/\nu$
- Turbulent kinetic energy, $k \sim 3u^2/2$
- Dissipation rate: $\varepsilon \sim u^3/\mathcal{L}$
- Kolmogorov scales:
  - ✓ Time: $\tau_K = (\nu/\varepsilon)^{1/2}$
  - ✓ Length: $\eta_K = (\nu^3/\varepsilon)^{1/4}$
  - ✓ Velocity: $\upsilon_K = (\nu\ \varepsilon)^{1/4}$

Questions & Discussion

1. What is the smallest length scale in turbulent flows?
2. How does the inertial subrange function in energy cascade?
3. The relation between the "Largest" and "Smallest" length scales in terms of the turbulent Reynolds number.
   (C.f. Slide 32)

Governing equations of fluid motions





- Mass conservation → **continuity equation**
- Momentum conservation → **momentum equations** (**Navier-Stokes** equations) for three velocity components
- For flow problems accompanied with heat / mass transfer → **heat or mass transport equation** (for thermal energy/temperature or for concentration of a substance/contaminant).
  - Scalar transport (passive quantities): one-way coupling with flow (follow the flow and not modify the flow motion)
  - Particle transport and deposition: One- or two-way coupling (interacting with the flow)
- For turbulent flows → **turbulent transport equation(s).**

Claude-Louis **Navier** & George Gabriel **Stokes**

Governing Equations of Fluid Motions

$$\frac{\partial Q}{\partial t}+\frac{\partial}{\partial x}\left(E-E_v\right)+\frac{\partial}{\partial y}\left(F-F_v\right)+\frac{\partial}{\partial z}\left(G-G_v\right)=0$$

$$Q=\begin{pmatrix}\rho\\ \rho u\\ \rho v\\ \rho w\\ \rho e_0\end{pmatrix}\quad E=\begin{pmatrix}\rho u\\ \rho u^2+p\\ \rho vu\\ \rho wu\\ \rho h_0 u\end{pmatrix}\quad F=\begin{pmatrix}\rho v\\ \rho uv\\ \rho v^2+p\\ \rho wv\\ \rho h_0 v\end{pmatrix}\quad G=\begin{pmatrix}\rho w\\ \rho uw\\ \rho vw\\ \rho w^2+p\\ \rho h_0 w\end{pmatrix}$$

$$E_v=\begin{pmatrix}0\\ \eta_{xx}\\ \eta_{yx}\\ \eta_{zx}\\ \eta_{xx}u+\eta_{yx}v+\eta_{zx}w-q_x\end{pmatrix}\quad F_v=\begin{pmatrix}0\\ \eta_{xy}\\ \eta_{yy}\\ \eta_{zy}\\ \eta_{xy}u+\eta_{yy}v+\eta_{zy}w-q_y\end{pmatrix}\quad G_v=\begin{pmatrix}0\\ \eta_{xz}\\ \eta_{yz}\\ \eta_{zz}\\ \eta_{xz}u+\eta_{yz}v+\eta_{zz}w-q_z\end{pmatrix}$$

Constitutive Relations

- To complete the Navier-Stokes equations
  - Newtonian fluid assumption (for most gases and liquids)

$$\eta_{ij} = \mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3}\frac{\partial u_k}{\partial x_k}\delta_{ij}\right) \qquad \mu = \text{dynamic molecular viscosity}$$

Note: for **incompressible flows**: $\dfrac{\partial u_k}{\partial x_k} = 0$

  - Fourier's heat law is assumed in most cases:

$$q_j = -\frac{c_p \mu}{\mathrm{Pr}}\frac{\partial T}{\partial x_j} \qquad \mathrm{Pr} = \text{Prandtl number}$$

.... for **incompressible flows** (constant density)

$$\frac{\partial u_i}{\partial x_i} = 0$$

$$\frac{\partial \rho u_i}{\partial t} + \frac{\partial}{\partial x_j}\left(\rho u_j u_i\right) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left(\mu\frac{\partial u_i}{\partial x_j}\right) + \rho g\beta\left(T - T_0\right)\delta_{i2}$$

$$\frac{\partial \rho T}{\partial t} + \frac{\partial}{\partial x_j}\left(\rho u_j T\right) = \frac{\partial}{\partial x_j}\left(\rho\alpha\frac{\partial T}{\partial x_j}\right) + S_T$$

$$\frac{\partial \rho C}{\partial t} + \frac{\partial}{\partial x_j}\left(\rho u_j C\right) = \frac{\partial}{\partial x_j}\left(\rho\gamma\frac{\partial C}{\partial x_j}\right) + S_C$$

- u_i – velocity component in x_i–direction(i=1,2,3, i.e. *u, v, w* in *x, y , z*-direction, resp.)
  T – temperature; C – concentration of mass (e.g. passive contaminants)
  ρ: density; g: gravitational acceleration (= 9.81 m/s^2); β: thermal expansion coefficient,
  T_0: reference temperature; α: thermal diffusivity; γ: mass-diffusion coefficient

- In 3D framework, the transport equation for an arbitrary flow quantity, Φ, can be cast in a general form:

$$\underbrace{\frac{\partial(\rho\Phi)}{\partial t}}_{\text{Transient term}}+\underbrace{\frac{\partial(\rho u\Phi)}{\partial x}+\frac{\partial(\rho v\Phi)}{\partial y}+\frac{\partial(\rho w\Phi)}{\partial z}}_{\text{Convection terms}}=$$

$$\underbrace{\frac{\partial}{\partial x}\left(\Gamma\frac{\partial\Phi}{\partial x}\right)+\frac{\partial}{\partial y}\left(\Gamma\frac{\partial\Phi}{\partial y}\right)+\frac{\partial}{\partial z}\left(\Gamma\frac{\partial\Phi}{\partial z}\right)}_{\text{Diffusion terms}}+\underbrace{S_\Phi}_{\text{Source term}}$$

Or, similarly:

$$\underbrace{\frac{\partial\rho\phi}{\partial t}}_{\text{Transient term}}+\underbrace{\nabla\cdot(\rho\vec{u}\phi)}_{\text{Convection term}}=\underbrace{\nabla\cdot(\Gamma\nabla\phi)}_{\text{Diffusion term}}+\underbrace{S_\phi}_{\text{Source term}}$$

$$\frac{\partial(\rho\Phi)}{\partial t}+\frac{\partial(\rho u\Phi)}{\partial x}+\frac{\partial(\rho v\Phi)}{\partial y}+\frac{\partial(\rho w\Phi)}{\partial z}=\frac{\partial}{\partial x}\left(\Gamma\frac{\partial\Phi}{\partial x}\right)+\frac{\partial}{\partial y}\left(\Gamma\frac{\partial\Phi}{\partial y}\right)+\frac{\partial}{\partial z}\left(\Gamma\frac{\partial\Phi}{\partial z}\right)+S_\Phi$$

- In conservative form, for an arbitrary variable, Φ,
  - Φ = 1 and S_Φ = 0→ continuity equation
  - Φ = u, v, w → momentum equations,
  - Φ = T (temperature), C (pollution concentration) → heat (T) or mass (C) transport equation
  - for turbulent quantities, e.g., k or ε, turbulent transport equation.
  - For turbulent flows → **turbulent transport equation(s).**
- **NOTE:**
  - For Φ = u, v or w: $S_\Phi=-\frac{\partial P}{\partial x}$ or $-\frac{\partial P}{\partial y}$ or $-\frac{\partial P}{\partial z}$
  - Γ is the diffusivity (or diffusion coefficient), for velocity, Γ = μ is the dynamic molecular viscosity, and ν = μ/ρ, which is the kinematic molecular viscosity.

For incompressible flows (density, ρ = constant), and in **2D** forms

Note that

$\nu = \mu/\rho$, which is the (constant) kinematic molecular viscosity, α and γ are, respectively, the (constant) thermal (heat) and mass diffusion coefficients

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0$$

$$\frac{\partial u}{\partial t}+\frac{\partial (uu)}{\partial x}+\frac{\partial (vu)}{\partial y}=-\frac{1}{\rho}\frac{\partial p}{\partial x}+\nu\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right)$$

$$\frac{\partial v}{\partial t}+\frac{\partial (uv)}{\partial x}+\frac{\partial (vv)}{\partial y}=-\frac{1}{\rho}\frac{\partial p}{\partial y}+\nu\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}\right)+g\beta(T-T_0)$$

Buoyancy force due to temperature difference

$$\frac{\partial T}{\partial t}+\frac{\partial (uT)}{\partial x}+\frac{\partial (vT)}{\partial y}=\alpha\left(\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}\right)+q_T$$

$$\frac{\partial C}{\partial t}+\frac{\partial (uC)}{\partial x}+\frac{\partial (vC)}{\partial y}=\gamma\left(\frac{\partial^2 C}{\partial x^2}+\frac{\partial^2 C}{\partial y^2}\right)+q_C$$

Simulation of turbulent flows in built environment

- Air flows created for built environment are usually **turbulent** and **incompressible**. In many cases, unless with large thermal variations, the *fluid properties* (e.g., density, molecular viscosity, heat or mass diffusivity …) are usually assumed *constant.*
- The eq. system is closed: 1 continuity eq. + 3 momentum eqs + 1 eq. for heat (*T*) and 1 eq. for mass (*C*, if pollution is concerned) transfer.

  Namely, 5 or 6 eqs for 5 or 6 unknown variables (*p, u, v, w* and *T* or *C*)
- Both laminar and turbulent flows are described/governed by these eqs
- In principle, solutions may be obtained with Direct Numerical Simulation (DNS) using the eq. system with no need of any model.
- For turbulent flows, however, DNS means to resolve all the turbulent scales (structures) down to the Kolmogorov scale, η_k, for which the required computing resources are usually unaffordable, or even become prohibitively high with the current computer capabilities …….

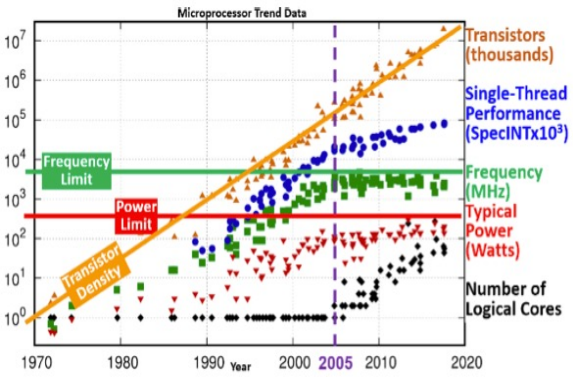
Turbulence Simulation Approaches

Increasing accuracy & comput. costs

- **Direct Numerical Simulation (DNS)**
  - Resolves all turbulence scales, no model
  - Produces accurate mean flow and resolves all-scale spectrum of turbulence
  - Requires massive amount of computer resources
- **Large Eddy Simulation (LES)**
  - Resolves large-scale turbulent motions
  - Produces mean flow and resolved large-scale turbulent quantities – effects of small scales modelled
  - Expensive, requires substantial computing resources – but less than DNS
- **Reynolds-Averaged Navier-Stokes Equations (RANS)**
  - Turbulent motions are "averaged" and fully modelled
  - Produces modelled turbulence and mean flow quantities
  - Spectral dynamics lost due to time-averaging
  - Cheap, commonly used in engineering computations

Increasing empiricism & comput. efficiency

Why not DNS?



- Note that
  $t/\tau \sim Re^{1/2}$, $\mathcal{L}/\eta \sim Re^{3/4}$
  & $\mathcal{L}/\lambda \sim Re^{1/2}$
  (λ: Taylor length scale)
- Grid nodes:
  $N_\eta \sim Re^{9/4}$ for DNS,
  $N_\lambda \sim Re^{3/2}$ for LES
- Time step:
  $N_{DNS} \sim N_{LES} \sim 100\ Re^{1/2}$
- Prohibitively demanding operations at large Re numbers

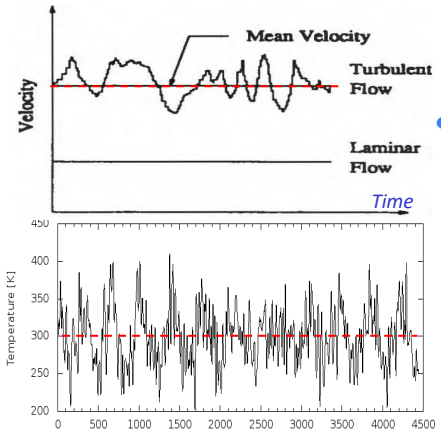
2050 or even later ready for DNS of flow over a full aircraft wing….?

For realistic applications, a model is used …

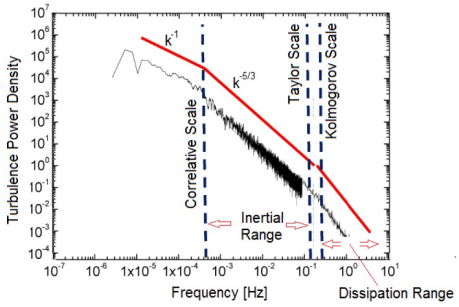
- Engineering applications are often characterized by complex geometries with complex flow phenomena. Computing resources for DNS may become prohibitively high, particularly for flows at large Re numbers
- The use of DNS is limited for low Re numbers fundamental flows with simple geometries
- Instead of solving all the turbulent structures, we solve the *mean flow* (by time-averaging) or the *large-scale structures* (by spatial filtering)
- Applying **time-averaging** (in RANS) or **spatial-filtering** (in LES) to the Navier-Stokes equations leads to *additional terms*, which need a model to close the eq. system.
  This leads to the so-called "**Closure Problem**"

→ Turbulence Modelling

Instantaneous *U(t) (velocity)* and *T(t)* (temperature)



Energy spectrum



Practically, mean flow properties are often required and of primary interest….

⇨Time-averaged process is invoked

Reynolds Averaging (Ensemble Averaging)

- An instantaneous flow quantity, *f*, is decomposed into a **mean** part, $\bar{f}$, and a **fluctuating** part, f':

$$f = \bar{f} + f'$$

With *N* identical experiments, ensemble averaging gives

$$\bar{f}(x,t) = \frac{1}{N}\sum_{1}^{N} f(x,t)$$

Or by time-averaging over a sufficient time period, T, with

$$\bar{f}(x,t) = \lim_{T\to\infty} \frac{1}{T}\int_{t_0}^{t_0+T} f(x,t)\,dt$$

Note: an overbar " ¯ " denotes Reynolds-averaged **mean** quantity

Reynolds Averaging

- Note for an arbitrary instantaneous flow quantity, f, its **mean** part, $\bar{f}$, and a **fluctuating** part, f', have the following relations, the overbar denotes Reynolds-averaging:

$$\overline{f'} = 0$$
$$\overline{\bar{f}f'} = 0$$
$$\overline{\bar{f}} = \bar{f}$$
$$\overline{\bar{f}\,\bar{g}} = \bar{f}\,\bar{g}$$

Example: Reynolds Averaging

$$u=\bar{u}+u',v=\bar{v}+v',w=\bar{w}+w',p=\bar{p}+p',T=\bar{T}+T',C=\bar{C}+C'$$

- For mass transfer in 1-dimensional (1D) incompressible flow (γ = constant – mass diffusivity), the instantaneous mass concentration is governed by

$$\frac{\partial C}{\partial t}+\frac{\partial(uC)}{\partial x}=\gamma\frac{\partial^2 C}{\partial x^2}$$

- Reynolds-averaging (*commute with differentiation*) leads to:

$$\overline{\frac{\partial C}{\partial t}}+\overline{\frac{\partial(uC)}{\partial x}}=\gamma\overline{\frac{\partial^2 C}{\partial x^2}}\Rightarrow\frac{\partial\bar{C}}{\partial t}+\frac{\partial\overline{(uC)}}{\partial x}=\gamma\frac{\partial^2\bar{C}}{\partial x^2}$$

$$\overline{(uC)}=\overline{(\bar{u}+u')(\bar{C}+C')}=\overline{(\bar{u}\bar{C}+2\bar{u}C'+u'C')}\Rightarrow\overline{uC}=\bar{u}\bar{C}+\overline{u'C'}$$

⇒ The Reynolds-Averaged equation

$$\frac{\partial\bar{C}}{\partial t}+\frac{\partial(\bar{u}\bar{C})}{\partial x}=\gamma\frac{\partial^2\bar{C}}{\partial x^2}-\frac{\partial(\overline{u'C'})}{\partial x}=\frac{\partial}{\partial x}\left(\gamma\frac{\partial\bar{C}}{\partial x}-\overline{u'C'}\right)$$

…. for **incompressible flows** (constant density)

- Applying Reynolds-averaging to the governing eqs yields

$$\frac{\partial\bar{u}_i}{\partial x_i}=0$$

$$\frac{\partial\bar{u}_i}{\partial t}+\frac{\partial}{\partial x_j}(\bar{u}_j\bar{u}_i)=-\frac{\partial\bar{p}}{\partial x_i}+\frac{\partial}{\partial x_j}\left(\nu\frac{\partial\bar{u}_i}{\partial x_j}+\tau_{ij}\right)+g\beta(\bar{T}-T_0)\delta_{i2}$$

$$\frac{\partial\bar{T}}{\partial t}+\frac{\partial}{\partial x_j}(\bar{u}_j\bar{T})=\frac{\partial}{\partial x_j}\left(\alpha\frac{\partial\bar{T}}{\partial x_j}+h_j\right)+q_T$$

$$\frac{\partial\bar{C}}{\partial t}+\frac{\partial}{\partial x_j}(\bar{u}_j\bar{C})=\frac{\partial}{\partial x_j}\left(\gamma\frac{\partial\bar{C}}{\partial x_j}+r_j\right)+q_C$$

Buoyancy force, usually = 0 for flows with no thermal effects

➢ **Unknown turbulent terms:** $\tau_{ij}=-\overline{u_i'u_j'}$, $h_j=-\overline{u_j'T'}$, $r_j=-\overline{u_j'C'}$

- Due to Reynolds-averaging for nonlinear terms, turbulence models needed for these unknown terms to close the equation system.

Exercise:

Using the equations on slide 43 and the relations on slides 49 and 50, try to derive the Reynolds-averaged equations for $\bar{u}$ and $\bar{v}$ in a two-dimensional framework and compare your derivation with these on slide 52 (the tensorial form).

For example, for the $\bar{u}$-equation in two-dimensional (2D) form:

$$\frac{\partial \bar{u}}{\partial t}+\frac{\partial\left(\bar{u}\,\bar{u}\right)}{\partial x}+\frac{\partial\left(\bar{u}\,\bar{v}\right)}{\partial y}=-\frac{1}{\rho}\frac{\partial \bar{p}}{\partial x}+\nu\left(\frac{\partial^2 \bar{u}}{\partial x^2}+\frac{\partial^2 \bar{u}}{\partial y^2}\right)+turb$$

(Note: an over-bar has been used for the time- or Reynolds-averaged quantities,)

The unknown term denoted "*turb*" is the turbulent stresses for the *u*-equation after time-averaging, **what are included in this term?** Similarly, think about the unknown for the $\bar{v}$-equation for a 2D case.

Favre Averaging

- For **compressible flows**, Reynolds-averaging yields unknown terms even in the continuity equation:

$$\frac{\partial \bar{\rho}}{\partial t}+\frac{\partial}{\partial x_i}\left(\bar{\rho}\bar{u}_i+\overline{\rho' u_i'}\right)=0$$

- Favre averaging (density-weighted time averaging) eliminates the density fluctuations from the averaged equations:

$$\tilde{f}=\frac{\overline{\rho f}}{\bar{\rho}}$$

  - Note: $f''=f-\tilde{f},\ \overline{\rho f''}=0,\ \tilde{f}''=0,\ \overline{f''}\neq 0$

Usually, incompressible flows in built environment …

The Bousinessq Approximation

- Transport of momentum by turbulence is a diffusive process and the Reynolds stresses can be modelled by an eddy-viscosity concept in analogy to its molecular counterpart

$$\tau_{ij} = -\overline{u_i' u_j'} = \nu_t \left(\frac{\partial \overline{u}_i}{\partial x_j} + \frac{\partial \overline{u}_j}{\partial x_i} \right) - \frac{2}{3} k \delta_{ij}$$

  - k is the turbulent kinetic energy,

$$k = \frac{1}{2} \overline{u_i' u_i'} = \frac{1}{2} \left(\overline{u_1'^2} + \overline{u_2'^2} + \overline{u_3'^2} \right) = \frac{1}{2} \left(\overline{u'^2} + \overline{v'^2} + \overline{w'^2} \right)$$

  - Note:
  - "*Isotropic*": equal eddy viscosity in all directions
  - "*Equilibrium alignment*": between stress tensor, τ_{ij}, and strain rate tensor, S_{ij}.
  - "*Analogy*": between turbulent eddies and molecules

Questions:

Based on the Bousinessq hypothesis, modelling the following turbulent stresses,

$-\overline{u'v'}$, $-\overline{u'u'}$ and $-\overline{v'v'}$

Referring to the derivation you made on slide 53, in which equation these turbulent stresses will be present, respectively?

Averaging the one-dimension transport equation for temperature *T*

$$\frac{\partial \overline{T}}{\partial t} + \frac{\partial}{\partial x}\left(\overline{uT}\right) = \frac{\partial}{\partial x}\left(D \frac{\partial \overline{T}}{\partial x} \right) \Rightarrow \frac{\partial \overline{T}}{\partial t} + \frac{\partial}{\partial x}\left(\overline{u}\,\overline{T}\right) = \frac{\partial}{\partial x}\left(D \frac{\partial \overline{T}}{\partial x} - \overline{u'T'} \right)$$

Try to reach the above derivation ……the second term in the parenthesis on the right-hand side is the turbulent heat-flux term which needs to be modelled. *D* is the heat diffusivity of the fluid.

… modelling of the turbulent eddy viscosity, ν_t, thus becomes essential !

- Similarly, the turbulent heat/concentration flux is modelled by an alignment between the flux and temperature/concentration gradient:

$$h_j^T = -\overline{u_j'T'} = \frac{\nu_t}{\mathrm{Pr}_t}\frac{\partial \bar{T}}{\partial x_j}, \qquad h_j^C = -\overline{u_j'C'} = \frac{\nu_t}{Sc_t}\frac{\partial \bar{C}}{\partial x_j}$$

- The "key" to **eddy-viscosity based modelling** is thus to formulate ν_t, which can be expressed, with dimensional analysis, as a product of a turbulence length scale, L, and turbulence velocity scale, u, namely, $\nu_t \sim uL$, or equivalently, in terms of L and a turbulence time scale, t, $\nu_t \sim L^2/t$.

  Pr_t (= 0.5-0.9) and Sc_t (= 0.7-1.0) are model constants, turbulent Prandtl or Schmidt numbers, respectively.

  - The "core" of RANS modelling is rooted in determining turbulent scales
  - The scales can be either expressed in empirical algebraic formulation or cast in transport equation(s)

  ➡ **Different "type" of eddy-viscosity based models**

Example: Standard k-ε model

$$\nu_t = C_\mu \frac{k^2}{\varepsilon} \quad \leftarrow \quad \nu_t \sim u\mathcal{L} \sim \sqrt{k}\,\frac{k^{3/2}}{\varepsilon}$$

$$\frac{\partial k}{\partial t} + \bar{u}_j \frac{\partial k}{\partial x_j} = P_k + \frac{\partial}{\partial x_j}\left(\left(\nu + \frac{\nu_t}{\sigma_k}\right)\frac{\partial k}{\partial x_j}\right) - \varepsilon$$

$$\frac{\partial \varepsilon}{\partial t} + \bar{u}_j \frac{\partial \varepsilon}{\partial x_j} = C_{1\varepsilon}\frac{\varepsilon}{k}P_k + \frac{\partial}{\partial x_j}\left(\left(\nu + \frac{\nu_t}{\sigma_\varepsilon}\right)\frac{\partial \varepsilon}{\partial x_j}\right) - C_{2\varepsilon}\frac{\varepsilon^2}{k}$$

- Note: in the k- and e equation, the term on the r.h.s. are respectively the production, diffusion and destruction term, $\nu_t = \mu_t/\rho$.

  σ's and C's are model constants, P_k is the production term of k

Example: Standard k-ω model

- The standard (STD) *k–ω* model (Wilcox, 1988)

$$\nu_t = C_\mu \frac{k}{\omega} \quad \longleftarrow \quad \boxed{\nu_t \sim u\mathcal{L} \sim \sqrt{k}\,\frac{k^{1/2}}{\omega}}$$

$$\frac{Dk}{Dt} = P_k + \frac{\partial}{\partial x_j}\left[\left(\nu + \frac{\nu_t}{\sigma_k}\right)\frac{\partial k}{\partial x_j}\right] - C_k k\omega$$

$$\frac{D\omega}{Dt} = C_{\omega 1}\frac{\omega}{k}P_k + \frac{\partial}{\partial x_j}\left[\left(\nu + \frac{\nu_t}{\sigma_\omega}\right)\frac{\partial \omega}{\partial x_j}\right] - C_{\omega 2}\omega^2$$

$$P_k = \tau_{ij}\frac{\partial u_i}{\partial x_j}, \tau_{ij} = -\overline{u_i u_j} = 2\nu_t\left(S_{ij} - \frac{\delta_{ij}}{3}S_{kk}\right) - \frac{2}{3}\delta_{ij}k$$

Some Basic Aspects of LES (Large Eddy Simulation)

- **Basic principle:**

⋆ **Spatial filtering**

$$\Rightarrow \begin{cases} \textit{energy-carrying large scales} \leftarrow \text{Resolved} \\ \textit{energy-scattering subgrid scales} \leftarrow \text{Modelled} \end{cases}$$

⋆ **Effects of subgrid scales are left in unknown SGS terms**

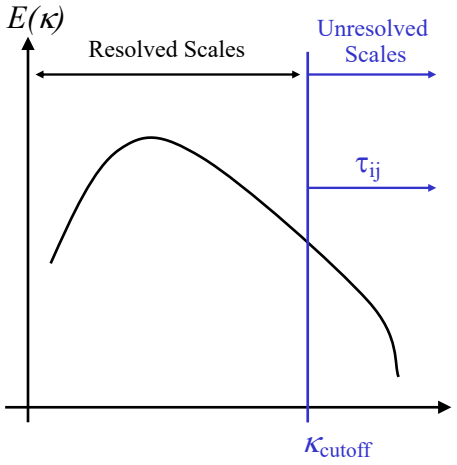
$\Rightarrow$ ***closure problem*** $\leftarrow$ **SGS modelling**

⋆ **Model-based simulation approach**

BUT *fewer modelling adjustments than RANS*

Filtering cut-off of energy cascade in LES

- LES resolves large, energy-bearing structures, but the effect of small, unresolved scales is modelled.
- The under-resolved scales are usually determined by the grid resolution, which is smaller than the grid size ⇨ SubGrid Scale.



Spatial-filtering Operation

◇ **The filtered, large-scale variable is defined by**

$$\bar{f}(\mathbf{x},t) = \int_D f(\mathbf{x}-\mathbf{x}',t)G(\mathbf{x}';\bar{\Delta})d\mathbf{x}',$$

where D is the entire flow domain, G is a low-pass filter function with a width of $\bar{\Delta}$ and

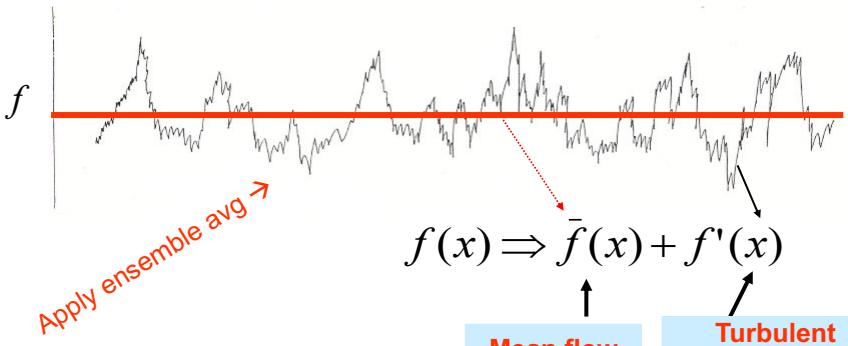
$$\int_D G(\mathbf{x}';\bar{\Delta})d\mathbf{x}' \equiv 1$$

⋆ **Positivity, strongly localized in space.**

⊛ **The instantaneous variable $f = \bar{f} + f''$, and**

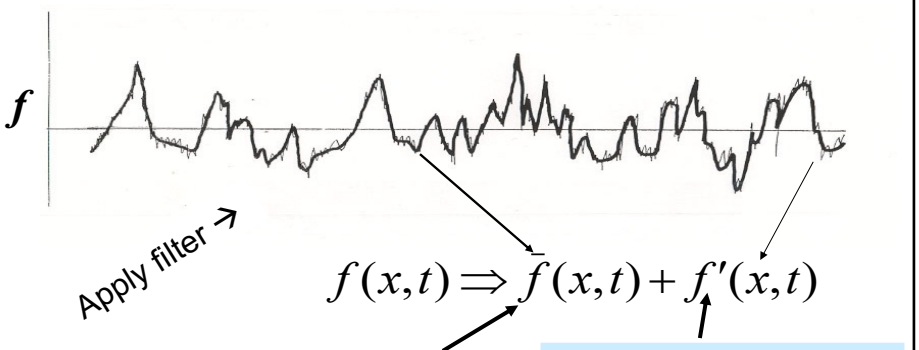
$$\overline{\frac{\partial f}{\partial x}} = \frac{\partial \bar{f}}{\partial x}, \quad \overline{\bar{f}} \neq \bar{f}, \quad \overline{f''} \neq 0, \quad (f - \overline{\bar{f}}) = (\overline{f''} + f'')$$

Effect of time-averaging



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Example: Spatial-Filtering Effect



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RANS *versus* LES

- RANS **models** "averaged turbulence", spectral effects are lost in averaging process.
- LES **resolves** large-scale turbulence with a hierarchy of scales down to the filtering cutoff (desirably in the inertial subrange).
- Sufficiently fine mesh required in all directions.
- Small-scale eddies tend to be more isotropic, **S**ub**G**rid-**S**cale (**SGS**) modelling may become more amenable and universal.
- In general LES is expected to be more accurate and reliable, but much more computationally costly.
- LES is always conducted in a 3D & time-dependent framework!

Machine Learning Incorporated in CFD

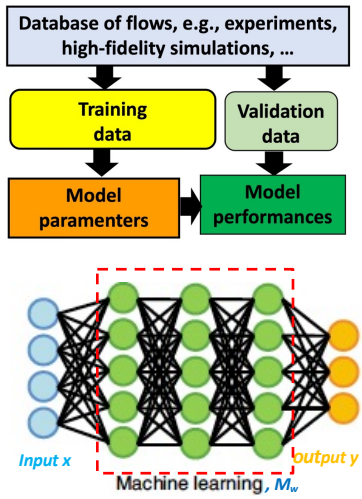
- Machine Learning (ML) involves **data-driven** learning processes that leverage existing data to develop models of making predictions or decisions about unknown data or events.
  - With a symbolic expression, I simplify ML as a process:

$$y = M_w(x)$$

where input x is mapped to output y via a parameterized model. The parameters, w, are learned through training by minimizing a loss (or cost) function $L(x, y)$, which measures prediction error.

The model learns "**what**" to represent **through latent space**, capturing essential features, and "**how**" to transform inputs **via learned operators** to achieve accurate predictions.

ML builds models from data using **optimization**.



Hybridization of ML and CFD

- Both ML and CFD may use "computer-based modelling", with "optimization" at their core.

Hybridization of "**data-driven**" ML and CFD, has shown great **possibilities**.

- Vast volume of flow data
- Increasing computational resources & capabilities
- Advanced algorithms
- Open-source software
- ……

Possibilities with ML in CFD:

- Acceleration of CFD simulations
- Reduced-Order Modelling (ROM)
- Turbulence closure modelling
- Flow control
- Aerodynamic optimization
- ……

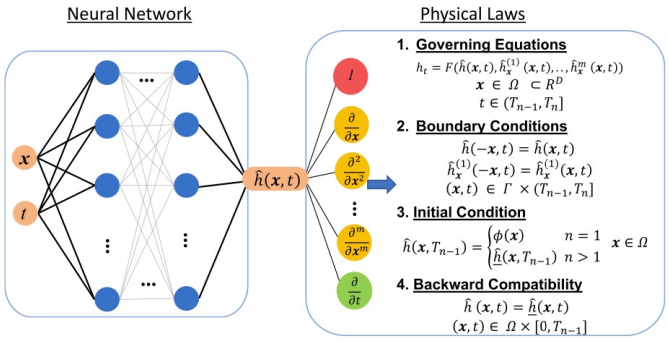
Challenges to address flow physics:

- High-dimensional freedoms
- Nonlinear interactions
- Multiple spatial and temporal scales
- Correlations in time and in space
- Interpretability
- ……

Machine Learning to Improve Turbulence Modelling

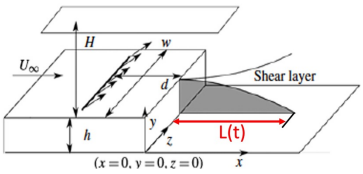
- Data-driven turbulence modelling (RANS and LES) has targeted, among others,
  - Optimized model coefficient or eddy viscosity for linear eddy viscosity-based model
  - Reynolds stress anisotropy tensor, $a_{ij} = \tau_{ij}/2k - 1/3\,\delta_{ij}$ for algebraic Reynolds stress models, $\mathbf{a} = \sum g_i \mathbf{T}^{(i)}$.
  - Improving modelling with physical symmetries, invariance, and in an interpretable framework.

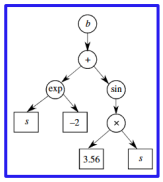
Physics-Informed Neural Networks (PINNs)

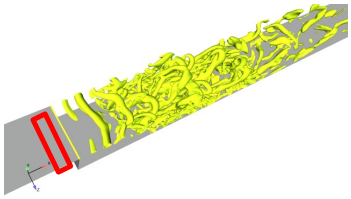


Machine Learning for Closed-Loop Flow Control

- Slotted jet with velocity U_j, as actuator to manipulate the recirculation area (or reattachment length, $L(t)$), of a backward-facing step (BFS) flow via two-dimensional real-time velocity field measurements (snapshots).
  - $L(t)$, normalized by its mean of the uncontrolled BFS flow, L_0, viz. $s(t) = L(t)/L_0$, as the sensor.
  - The control value, $b = U_j/U_{max}$, as a function of s, $b = K(s)$. U_{max}, is the max jet velocity.
  - Genetic Programming (GP) is deployed to find the control law optimizing a cost function, $J(s, b)$, where b is incorporated as a penalization by the actuation energy.





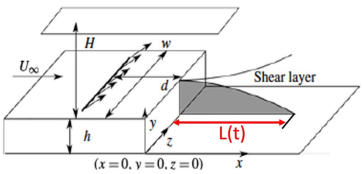


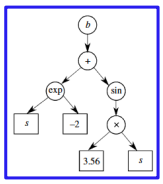
Gautier et al., *J. Fluid Mechanics*. 2015. 770, pp.442-457

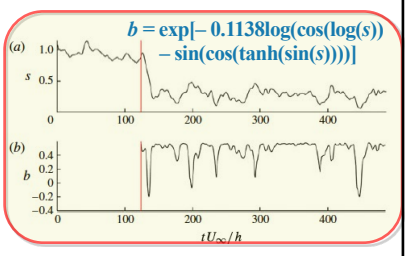
Conceptual Introduction on
Computational Fluid Dynamics (CFD) for Built Environment

Machine Learning for Closed-Loop Flow Control

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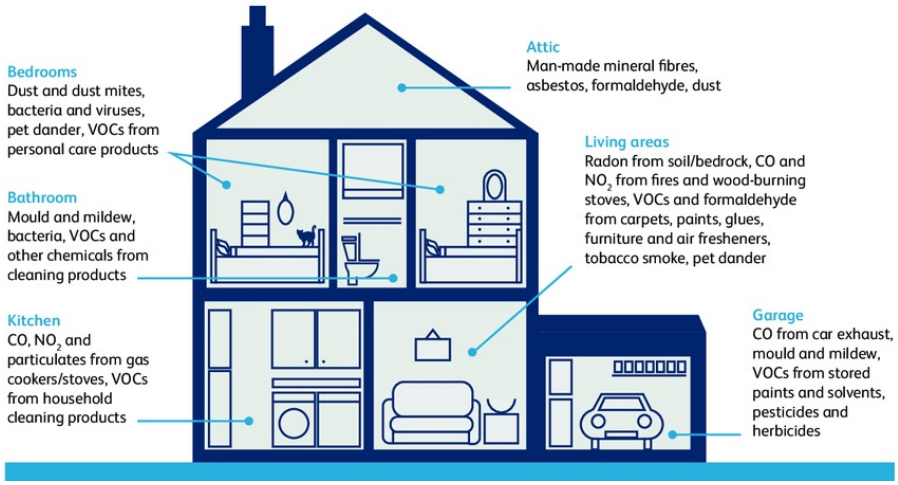




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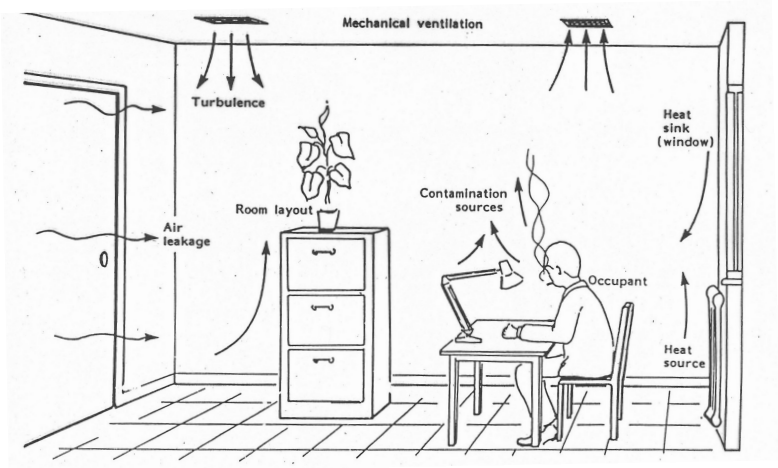
Turbulent airflow motions in built environment

- Most of flows present in built environment are turbulent and, in many cases, at low Re numbers, being accompanied with heat and mass transfer processes …
- Turbulence is responsible for, usually,
  - Increasing heat transfer, drag
  - Increasing pollutant release from surface
  - Increasing dispersion/transport of heat/contaminants in space
  - Potential flow-induced noise generation
  - Thermal discomfort (increasing draught risk …)
- Turbulence modeling is one of the major uncertainties in CFD analysis of built environment
- Appropriately accommodating turbulence effects in CFD of built environment is important for accurate predictions of air flow and heat/mass transfer



Possible sources of indoor pollutants

Typical factors to consider in CFD simulations of indoor air flow for accurate predictions

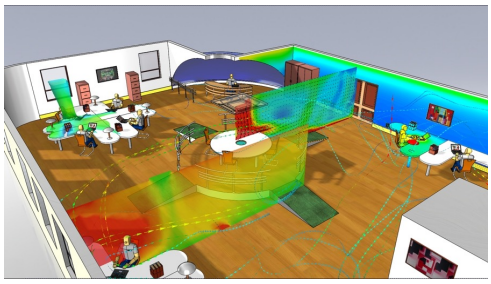


Procedure of CFD analysis

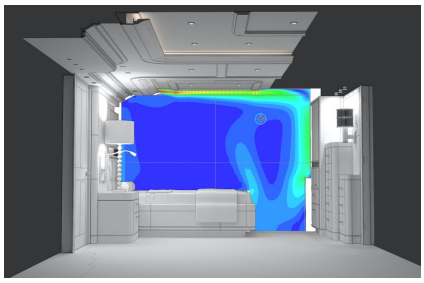
- CFD analysis is able to generates detailed predictions of indoor air freshness, building energy efficiency, occupant's thermal comfort, …
- **General procedure:**
  - Problem description (objectives, key flow, heat/mass-transfer physics, results expected from CFD predictions …)
  - Setup of computational domain & boundary conditions
  - Grid generation
  - Selection of numerical schemes & solution algorithms
  - Selection of models of flow physics (turbulence, particles…)
  - Preliminary studies of appropriateness on numerical setup
  - Further improvement of solution (e.g., using high-order numerical schemes, more realistic boundary conditions, refined grid resolution, more relevant turbulence models …)
  - Validation & verification
  - Analysis & interpretation of results (post-processing & visualization …)
  - Reporting of results and analysis

Typical CFD applications to built environment & indoor climate

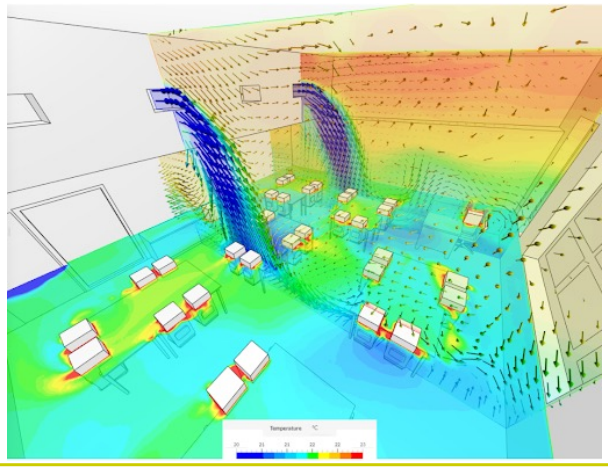
Ventilation in office room



Ventilation in bedroom



CFD analysis of airflow and thermal fields in a ventilated classroom – How would you do ….?



How to Verify, Validate, and Report Indoor Environment Modeling CFD Analyses

(A reference to read … …)

ASHRAE RP-1133

The CFD model should also be constructed in progressive stages. The sequence should be (1) correctly set up the flow domain and geometry, (2) progressively increase the complexity of the flow and thermal boundary conditions, and (3) then fine tune numerical schemes and convergence criteria.

CFD user plays an important role in the verification and validation. An accurate and successful simulation is a combined effort of the CFD code and the CFD user.

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CFD Analysis of Indoor Environment

4. CONCLUSIONS

A manual has been developed that describes how to verify, validate, and report indoor environment modeling CFD analyses. Verification involves identifying the relevant physical phenomena for an indoor environmental analysis and assessing whether or not a particular CFD code itself has the capability of accounting for those physical phenomena. Validation involves demonstrating the coupled ability of a user and a CFD code to accurately conduct a simulation of a representative indoor environment. Reporting results involves summarizing the CFD analysis so that others can make informed assessments of the value and quality of the CFD. This manual provides a set of instructions on how best to perform these three steps for a CFD analysis.

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The manual recommends verifying a CFD code for the modeling of the physical phenomena in an indoor environment based on the following aspects:

- Basic flow and heat transfer features Boundary conditions
- Turbulence models
- Auxiliary heat transfer and flow models
- Numerical methods
- Assessing CFD predictions

Verification is recommended for both developers and users of CFD codes. The developers can use the verification to check the code for bugs, while the users can become familiar with the features and capabilities of the code. Verification should be performed for simple cases with only one or two flow and heat transfer features of the physical phenomena of an indoor environment system.

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Validation is applied for:

- Confirming the abilities of the turbulence model and other auxiliary models at predicting all physical phenomena in a particular indoor environment, in order to apply the models for design and evaluation of a similar indoor environment category
- Confirming the discretization method, grid resolution, and numerical algorithm for the flow simulation
- Confirming the user's ability to use the CFD code to perform an indoor environment analysis

The validation covers the same aspects as those for verification. However, the validation is for a complete indoor environment system or several subsystems that represent the complete flow, heat and mass transfer features in an indoor environment. Although the validation is for a complete indoor environment system or several subsystems, the manual recommends a progressive validation procedure, because reliability is improved:

- For simple geometry, rather than a complicated one
- For convection, rather than combined convection, conduction, and radiation
- For single-phase flows, rather than multi-phase flows

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Although the format for reporting of CFD analysis does not necessarily have to be the same, the report should include the following for technical readers:

- Experimental design
- Turbulence models and auxiliary heat transfer and flow models
- Boundary conditions
- Numerical methods
- Assessing CFD predictions
- Drawing conclusions

The report should provide adequate analysis of the results, and clearly state the uncertainties and errors of the experimental data, if they are known. The report should also conclude the most important findings of the CFD analysis. The report may present lessons learned during the CFD analysis, and recommend measures for further improvements.

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The CFD model should also be constructed in progressive stages. The sequence should be (1) correctly set up the flow domain and geometry, (2) progressively increase the complexity of the flow and thermal boundary conditions, and (3) then fine tune numerical schemes and convergence criteria.

CFD user plays an important role in the verification and validation. An accurate and successful simulation is a combined effort of the CFD code and the CFD user.

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Review the notes for improved understandings…

- Review the lecture notes and try to understand the concepts/description of, among others,
  - What CFD means and its use in analyzing fluid flow and heat/mass transfer
  - What the three primary ingredients are for CFD simulations
  - Turbulence, turbulence scales
  - Why not DNS?
  - Why is turbulence modelling required?
  - What does Reynolds-averaging mean?
  - What is the difference between RANS and LES?
  - Fundamental issues in turbulence modelling based on the Bousinessq approximation .
- CFD analysis of built environment (cf also the distributed document)
  - The general procedure of CFD analysis
  - What validation and verification mean?
  - How the CFD prediction is assessed?
- **Review the questions/discussion/exercises given on slides 17, 30, 34, 35, 36, 53, 56.**

For any questions/suggestions on this lecture, write to

shpeng@kth.se or
peng@chalmers.se